

CONSULTANT'S FORUM

Sample Size/Power Calculation for Case–Cohort Studies

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SUMMARY. In epidemiologic studies and disease prevention trials, interest often involves estimation of the relationship between some disease endpoints and individual exposure. In some studies, due to the rarity of the disease and the cost in collecting the exposure information for the entire cohort, a case–cohort design, which consists of a small random sample of the whole cohort and all the diseased subjects, is often used. Previous work has focused on analyzing data from the case–cohort design and few have discussed the sample size issues. In this article, we describe two tests for the case–cohort design, which can be treated as a natural generalization of log-rank test in the full cohort design. We derive an explicit form for power/sample size calculation based on these two tests. A number of simulation studies have been used to illustrate the efficiency of the tests for the case–cohort design. An example is provided on how to use the formula.

KEY WORDS: Case–cohort design; Log-rank test.

1. Introduction

In epidemiologic cohort studies and disease prevention trials, it is often of interest to study the relationship between the exposure of interest and some disease endpoints. When only a very small fraction of the full cohort develops diseases in the given study time frame and the exposure measurement of interest is expensive to obtain, it can be very costly to obtain the exposure measurement for the entire cohort. The case–cohort design (Prentice, 1986) was proposed as a cost-effective alternative for such studies. In the case–cohort design, a simple random sample (subcohort) is drawn from the full cohort and the case–cohort sample consists of the subcohort and those subjects whose disease occurred during the study period.

Since its proposal, the case–cohort design has been used in many studies (e.g., Schouten et al., 1993; Liao et al., 1997; Savitz et al., 2000; Folsom et al., 2001) such as the Atherosclerosis Risk in Communities study (ARIC Investigators, 1989) described in more detail in Section 4. The method development for the case–cohort design has been focused on statistical inference for the Cox regression model. Prentice (1986) proposed a pseudo-likelihood approach where the risk set at each failure consisted of only those subjects who were at risk from the subcohort. Self and Prentice (1988) used another slightly modified pseudo-likelihood approach where the risk set at each failure consisted of those subjects who were at risk from the subcohort and the subject who failed at that time. They developed the asymptotic distribution theory for

the parameter estimator and a similar result was given for a more general sampling frame by Borgan, Goldstein, and Langholz (1995). Lin and Ying (1993) viewed the case–cohort design as a special case of the missing data problem and proposed an estimating equation approach for parameter estimation. They showed that their estimator is asymptotically normally distributed with a sandwich type of variance. Barlow (1994) proposed a jackknife method for estimating the variance of the parameter estimates. Chen and Lo (1999) further improved the estimation in Prentice (1986) by using different estimation of population distribution. Recently, Chen (2001a,b) considered the estimation in a proportional odds model and a semiparametric transformation model under such design.

Although there has been considerable development in the analysis of data from case–cohort studies, design issues have not been addressed. Sample size/power calculation is an important component at the design stage of a study. Specifically, for the case–cohort studies, it is important to determine the size of the subcohort needed for the study. Given a certain budget for a study, it is also of interest to know how much more power may be gained by using a case–cohort design, compared with a simple random sample design.

In this article, we present methods for computing sample size and power for case–cohort studies. Here the term “sample size” refers to the size of the subcohort. Specifically, we describe two log-rank type tests for a case–cohort design and derive the power and sample size formulas based on these tests.

A number of simulations are performed to compare the power of our proposed tests with the log-rank test using the full cohort and the subcohort to examine the relative efficiency.

2. Proposed Tests and Power Formulas

2.1 Proposed Tests

Assume that there are two groups and there are n_j subjects in group j ($j = 1, 2$). Let T_{ij} and C_{ij} denote the potential failure and censoring time for subject i in group j ($j = 1, 2; i = 1, \dots, n_j$) and they are assumed to be independent. Let $X_{ij} = T_{ij} \wedge C_{ij}$ denote the observed time and $\Delta_{ij} = I(T_{ij} \leq C_{ij})$ the failure indicator variable. Assume that the full cohort size is $n = n_1 + n_2$ and the subcohort size is $\tilde{n}$. Let ξ_{ij} be the indicator that subject i of group j is selected into the subcohort. Our main goal is to compare the hazard rates between the two groups; or equivalently, to test the null hypothesis $H_0: \Lambda_1(t) = \Lambda_2(t)$, $t \in [0, \tau]$, where $\Lambda_j(t)$ is the cumulative hazard function of the failure time in group j ($j = 1, 2$).

If all data from the full cohort were available, one could use the log-rank test in comparing the hazard rates between the two groups. Let $\tilde{Y}_j(t)$ be the number of subjects at risk at time t in the j th group and let $w(t)$ be the weight function. Then, the log-rank test is given by

$$\sum_{i=1}^{n_1} \frac{\Delta_{i1} w(X_{i1}) \tilde{Y}_2(X_{i1})}{\tilde{Y}_1(X_{i1}) + \tilde{Y}_2(X_{i1})} - \sum_{i=1}^{n_2} \frac{\Delta_{i2} w(X_{i2}) \tilde{Y}_1(X_{i2})}{\tilde{Y}_1(X_{i2}) + \tilde{Y}_2(X_{i2})}. \quad (1)$$

However, with a case-cohort design, the above statistic is not directly applicable since we only know the number of subjects at risk for group j in the subcohort, which is denoted by $\tilde{Y}_j(t)$. If the sampling fraction for the subcohort is q , then it is natural to approximate $\tilde{Y}_j(t)$ by $\tilde{Y}_j(t)/q$. By using this approximation in (1), we obtain our first test statistic

$$\sum_{i=1}^{n_1} \frac{\Delta_{i1} w(X_{i1}) \tilde{Y}_2(X_{i1})}{\tilde{Y}_1(X_{i1}) + \tilde{Y}_2(X_{i1})} - \sum_{i=1}^{n_2} \frac{\Delta_{i2} w(X_{i2}) \tilde{Y}_1(X_{i2})}{\tilde{Y}_1(X_{i2}) + \tilde{Y}_2(X_{i2})}. \quad (2)$$

In a full cohort design, the log-rank test is known to be the same as the score function of the Cox partial likelihood function under the null hypothesis. Thus, it is not surprising to see that the above test statistic (2) for the case-cohort design is exactly the score function of the pseudopartial likelihood function given by Self and Prentice (1988), where the risk set at each failure time consists of those subjects who were at risk from the subcohort and the subject who failed at that time. As shown in Self and Prentice (1988), (2) has an asymptotic normal distribution. Then a standardized version of (2) can be used to test our null hypothesis and we denote such a test as SP_n .

In addition, we can propose another test statistic by using a different way to approximate $\tilde{Y}_j(t)$ in (1) as follows: with a case-cohort design, each nonfailure subject in the subcohort represents $n/\tilde{n}$ nonfailure subjects in the full cohort. Therefore, the number of subjects at risk for the full cohort can be approximated by the sum of the number of subjects at risk who failed by the end of the study and $n/\tilde{n}$ times the number of subjects at risk in the subcohort who did not fail by the

end of the study. That is, we can approximate the risk set of the full cohort at time t by

$$Y_j^*(t) = \frac{n}{\tilde{n}} \sum_{i=1}^{n_j} I(X_{ij} \geq t, \xi_{ij} = 1, \Delta_{ij} = 0) + \sum_{i=1}^{n_j} I(X_{ij} \geq t, \Delta_{ij} = 1), \quad j = 1, 2.$$

Substituting this approximation into the usual log-rank test statistic (1), we obtain another log-rank type test statistic

$$\sum_{i=1}^{n_1} \frac{\Delta_{i1} w(X_{i1}) Y_2^*(X_{i1})}{Y_1^*(X_{i1}) + Y_2^*(X_{i1})} - \sum_{i=1}^{n_2} \frac{\Delta_{i2} w(X_{i2}) Y_1^*(X_{i2})}{Y_1^*(X_{i2}) + Y_2^*(X_{i2})} \quad (3)$$

and we denote it as V_n .

Essentially, both log-rank type tests for the case-cohort design are to approximate the risk sets in the full cohort with the observed risk set. Due to this approximation, it is natural that the tests for the case-cohort design are less efficient than the test when the full cohort is available. The loss of efficiency will be investigated in Section 3.

2.2 Power Formulas Based on the Test Statistics SP_n and V_n

From the result in Self and Prentice (1988), when n_j/n approximates some $p_j \in (0, 1)$ and $\tilde{n}/n$ converges to q , $n^{-1/2} SP_n$ has an asymptotic normal distribution under H_0 . Furthermore, its asymptotic variance can be consistently estimated by the following formula:

$$\begin{aligned} \hat{\sigma}_{sp}^2 &= \hat{\sigma}^2 + \frac{2(1-\hat{q})}{n} \\ &\times \sum_{j=1}^2 \sum_{i=1}^{n_j} \left\{ \frac{\Delta_{ij} w(X_{ij}) \tilde{Y}_1(X_{ij}) \tilde{Y}_2(X_{ij})}{(\tilde{Y}_1(X_{ij}) + \tilde{Y}_2(X_{ij}))^2} \right. \\ &\quad \times \left[\sum_{j'=1}^2 \sum_{i'=1}^{n_{j'}} \frac{\Delta_{i'j'} w(X_{i'j'}) I(X_{i'j'} \leq X_{ij})}{\tilde{Y}_1(X_{i'j'}) + \tilde{Y}_2(X_{i'j'})} \right] \Big\} \\ &- \frac{(1-\hat{q})}{n} \sum_{j=1}^2 \sum_{i=1}^{n_j} \frac{\Delta_{ij} w(X_{ij}) \tilde{Y}_1(X_{ij}) \tilde{Y}_2(X_{ij})}{(\tilde{Y}_1(X_{ij}) + \tilde{Y}_2(X_{ij}))^3}, \end{aligned} \quad (4)$$

where $\hat{q} = \tilde{n}/n$, $\hat{\sigma}^2$ is a consistent estimate for the asymptotic variance of the log-rank test in the full cohort and is given by

$$\begin{aligned} \hat{\sigma}^2 &= \frac{1}{n} \left\{ \sum_{i=1}^{n_1} \frac{\Delta_{i1} w(X_{i1}) \tilde{Y}_2(X_{i1})^2}{(\tilde{Y}_1(X_{i1}) + \tilde{Y}_2(X_{i1}))^2} \right. \\ &\quad \left. + \sum_{i=1}^{n_2} \frac{\Delta_{i2} w(X_{i2}) \tilde{Y}_1(X_{i2})^2}{(\tilde{Y}_1(X_{i2}) + \tilde{Y}_2(X_{i2}))^2} \right\}. \end{aligned}$$

Hence, at the significance level α , to test $H_0: \Lambda_1(t) = \Lambda_2(t)$, $t \in [0, \tau]$ versus $H_A: \Lambda_1(t) < \Lambda_2(t)$, $t \in [0, \tau]$, we reject H_0 (one-sided test) if $n^{-1/2} SP_n / (\hat{\sigma}_{sp}^2)^{1/2} < Z_\alpha$, where Z_α denotes the α th quantile of standard normal distribution.

To provide simple power and sample size formula, we consider the case for $w(t) = 1$. In addition, we assume that

- The censoring distributions are the same in the two groups.
- The number of failures is very small in the full cohort but much larger than one.
- No ties of failures are observed.

Under these conditions, our sample size calculation is under the alternative hypothesis

$$H_A : \Lambda_2(t) = e^\theta \Lambda_1(t),$$

where $\theta = O(\tilde{n}^{-1/2})$. The Appendix gives the simplification details. It turns out that the power of our test at H_A can be approximated by

$$\Phi \left(Z_\alpha + \tilde{n}^{1/2} \theta \sqrt{\frac{p_1 p_2 p_D}{q + (1-q)p_D}} \right), \quad (5)$$

where α is the significance level, θ is the log-hazard ratio for two groups, p_j , $j = 1, 2$, are the proportion of the two groups in the population ($p_1 + p_2 = 1$), $\tilde{n}$ is the total number of subjects in the subcohort, p_D is the proportion of the failures in the full cohort, and q is the sampling fraction of the subcohort. Obviously, since $q \leq (q + (1-q)p_D) \leq 1$, the power from our test statistic is between $\Phi(Z_\alpha + \tilde{n}^{1/2} \theta (p_1 p_2 p_D)^{1/2})$ and $\Phi(Z_\alpha + n^{1/2} \theta (p_1 p_2 p_D)^{1/2})$. The former is the power of the log-rank test which only uses the subcohort observations, while the latter is the power of the log-rank test which uses the full cohort observations. We thus conclude that

$$\mathcal{P}_s \leq \mathcal{P}_{sp} \leq \mathcal{P}_f,$$

where $\mathcal{P}_s$ is the power of the log-rank test using the subcohort observations, $\mathcal{P}_{sp}$ is the power based on the test statistic SP_n , and $\mathcal{P}_f$ is the power of the log-rank test using the full-cohort observations.

Similarly, we can show that $n^{-1/2}V_n$ has an asymptotic normal distribution under H_0 . However, its asymptotic variance has a more complicated form than that of $n^{-1/2}SP_n$. A consistent estimate of the asymptotic variance, $\hat{\sigma}_V^2$, is given in the Appendix. Hence, at the significance level α , to test $H_0 : \Lambda_1(t) = \Lambda_2(t)$, $t \in [0, \tau]$ versus $H_A : \Lambda_1(t) < \Lambda_2(t)$, $t \in [0, \tau]$, we reject H_0 (one-sided test) if $n^{-1/2}V_n/(\hat{\sigma}_V^2)^{1/2} < Z_\alpha$. Interestingly, the power formula for this test statistic under the simplification conditions (a)–(c) for the local alternative hypothesis $H_A : \Lambda_2(t) = e^\theta \Lambda_1(t)$, where $\theta = O(\tilde{n}^{-1/2})$, is the same as the approximated power formula based on SP_n in (5).

3. Numerical Results

By using the power formula derived in the previous section, we present in Table 1 the results of the theoretical power calculation of the log-rank test using the full cohort, the tests using SP_n and V_n , and the log-rank test using the subcohort, which are indicated in Table 1 by $\mathcal{P}_f$, $\mathcal{P}_{sp}$, $\mathcal{P}_V$, and $\mathcal{P}_s$, respectively. The calculation is based on the formula in (5) and is

Table 1
Theoretical power calculation for case-cohort design ($\alpha = 0.05$)

Full cohort size n	Failure prop. p_D (%)	Sampling prop.					
		p_1	θ	q (%)	$\mathcal{P}_f$	$\mathcal{P}_{sp} = \mathcal{P}_V$	$\mathcal{P}_s$
1000	10	0.3	0.5	10	0.741	0.507	0.179
				20	0.741	0.615	0.267
			1.0	10	0.998	0.953	0.422
				20	0.998	0.987	0.657
		0.5	0.5	10	0.804	0.567	0.196
				20	0.804	0.680	0.299
			1.0	10	1.000	0.976	0.475
				20	1.000	0.995	0.723
		5	0.3	10	0.490	0.382	0.129
				20	0.490	0.434	0.179
			1.0	10	0.945	0.852	0.267
				20	0.945	0.905	0.422
		0.5	0.5	10	0.549	0.430	0.139
				20	0.549	0.487	0.196
			1.0	10	0.971	0.902	0.299
				20	0.971	0.943	0.475
5000	5	0.3	0.5	1	0.976	0.437	0.100
				2	0.976	0.620	0.129
			1.0	1	1.000	0.907	0.179
				2	1.000	0.988	0.267
		0.5	0.5	1	0.989	0.490	0.106
				2	0.989	0.685	0.139
			1.0	1	1.000	0.945	0.196
				2	1.000	0.995	0.299
		1	0.3	1	0.490	0.310	0.069
				2	0.490	0.375	0.078
			1.0	1	0.945	0.743	0.093
				2	0.945	0.844	0.118
		0.5	0.5	1	0.549	0.348	0.071
				2	0.549	0.422	0.081
			1.0	1	0.971	0.805	0.098
				2	0.971	0.895	0.126

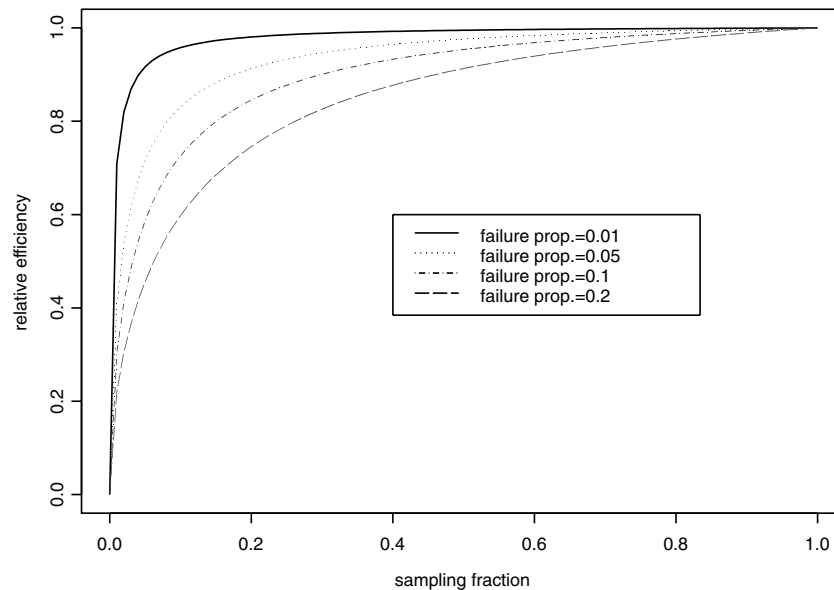


Figure 1. Relative efficiency of the case-cohort design relative to the full cohort.

done for different full cohort size (n), the failure rate (p_D), the fraction of the first group (p_1), the log-hazard rate (θ), and the sampling fraction (q) for the significance level 0.05. It is noted that since under assumptions (a)–(c) the variances of SP_n and V_n are asymptotically equal, all the numerical and theoretical results of these two tests are the same. Thus, the results for SP_n and V_n are reported in one single column in Table 1. From Table 1, we can see that the tests for the case-cohort design are always much more powerful than the subcohort analysis when the failure rate is low. As expected, the power increases with the sampling fraction. Under some circumstances, even a small sampling fraction (10% or 20%) can give almost as much power as the log-rank test using the full cohort. We also plot the relative efficiencies (ratio of the asymptotic variances) of the case-cohort design relative to the full cohort analysis over different sampling fractions (q) for certain failure rates (p_D) in Figure 1. We can see that when the failure rate is as low as 5%, the relative efficiency of the case-cohort analysis is close to 1; the relative efficiency of the case-cohort analysis becomes smaller when the failure rate is larger.

Next, simulation studies are performed to examine the type I error rates of the tests and to compare the empirical power of four tests: the log-rank test using the full cohort (T_{full}), the test statistics SP_n and V_n , and the log-rank test using the subcohort (T_{sub}). The failure time is generated from exponential distribution with constant rate λ_j for group j ($j = 1, 2$) and the censoring time is generated from a uniform distribution between $[0, \tau]$. We choose $\lambda_1 = \lambda_2 = 0.1$ to study the empirical type I error rate and choose $\lambda_1 = 0.1$ and $\lambda_2 = 0.15$ or 0.2 for the power comparison. Moreover, we vary the choice of τ to obtain different censoring proportions with $\tau = 2.15, 1.05$, and 0.23 corresponding to 90%, 95%, and 99% censoring for the first group, respectively. Each simulation is repeated 1000 times. We consider different combinations of

full cohort size ($n = 1000, 5000$), group fraction ($p_1 = 0.3, 0.5$), censoring proportion in the first group ($1 - p_D = 90\%, 95\%, 99\%$), and sampling fraction ($q = 0.1, 0.2$). Table 2 gives the empirical type I error rates of all four tests when $\lambda_1 = \lambda_2 = 0.1$. We can see that the type I error rates of the two tests are reasonably good for a large cohort with very few failures. Comparatively, the log-rank test based on the subcohort may produce either too large or too small type I errors due to the small number of failures in the subcohort. Table 3 presents the empirical power of all four tests when $\lambda_1 = 0.1$ and $\lambda_2 = 0.15$ and 0.2 . The results indicate that when the failure rate is low in a large cohort, both of the test statistics SP_n and V_n give similar power under the alternative hypothesis. Moreover, their power decreases as the censoring proportion increases and the power increases as the sampling fraction increases.

Based on the reviewers' suggestion, we also compared the performance of SP_n and V_n when event rate is not rare. We perform a simulation study with the event rate of 25% and the size of the full cohort of 200 or 400. The results are summarized in Table 4. From the table, we observe that when the event rate is not small, V_n tends to provide larger power than SP_n and the difference can be as much as 6%. This makes sense since V_n uses more information in approximating the risk set of the full cohort.

One may also be interested in the relationship between the testing power and the sampling proportion given a full cohort size. In Figure 2, we plot the ratio of the power of the case-cohort design to the power of the full cohort test versus the sampling fraction, where the full cohort size is 1000 or 5000. We choose $p_1 = p_2 = 0.5$, $\theta = 0.5$ or 1 , and $p_D = 0.1$ or 0.2 . From the plot, we see that when the full cohort size is large and the failure rate is low, a very small sampling proportion can provide similar power as using the whole cohort.

Table 2
Simulated type I errors for the four log-rank tests ($\lambda_1 = \lambda_2 = 0.1$, $\alpha = 0.05$)

n	p_1	Censoring prop. $1 - p_D$ (%)	Sampling prop. q (%)	T_{full}	SP_n	V_n	T_{sub}
1000	0.3	90	10	0.061	0.044	0.044	0.089
			20	0.061	0.044	0.045	0.059
		95	10	0.076	0.051	0.052	0.139
			20	0.076	0.072	0.071	0.093
	0.5	90	10	0.049	0.043	0.040	0.041
			20	0.049	0.046	0.043	0.046
		95	10	0.044	0.042	0.042	0.042
			20	0.044	0.051	0.050	0.049
5000	0.3	95	1	0.066	0.039	0.042	0.124
			2	0.066	0.043	0.047	0.153
		99	1	0.061	0.042	0.042	0.009
			2	0.061	0.046	0.046	0.059
	0.5	95	1	0.054	0.057	0.057	0.039
			2	0.054	0.057	0.062	0.061
		99	1	0.034	0.046	0.048	0.004
			2	0.034	0.036	0.038	0.031

Table 3
Simulated test powers for the four log-rank tests with rare events ($\lambda_1 = 0.1$, $\alpha = 0.05$)

λ_2	n	p_1	Censoring prop. $1 - p_D$ (%)	Sampling prop. q (%)	T_{full}	SP_n	V_n	T_{sub}
0.15	1000	0.3	90	10	0.686	0.334	0.340	0.211
				20	0.686	0.463	0.469	0.269
			95	10	0.457	0.280	0.278	0.217
				20	0.457	0.357	0.357	0.235
		0.5	90	10	0.729	0.376	0.378	0.145
				20	0.729	0.517	0.523	0.247
			95	10	0.455	0.303	0.308	0.112
				20	0.455	0.362	0.358	0.161
	5000	0.3	95	1	0.950	0.238	0.238	0.238
				2	0.950	0.369	0.377	0.216
			99	1	0.515	0.197	0.204	0.051
				2	0.515	0.285	0.289	0.103
		0.5	95	1	0.970	0.320	0.330	0.094
				2	0.970	0.441	0.458	0.125
0.2	1000	0.3	90	1	0.532	0.268	0.268	0.020
				2	0.532	0.332	0.318	0.046
			95	1	0.984	0.669	0.680	0.370
				20	0.984	0.845	0.866	0.568
		0.5	90	10	0.874	0.595	0.605	0.300
				20	0.874	0.723	0.724	0.400
			95	10	0.992	0.775	0.796	0.329
				20	0.992	0.910	0.912	0.562
	5000	0.3	90	10	0.901	0.670	0.680	0.220
				20	0.901	0.783	0.787	0.342
			95	1	1.000	0.485	0.495	0.317
				2	1.000	0.728	0.730	0.288
		0.5	90	1	0.906	0.433	0.441	0.077
				2	0.906	0.585	0.588	0.176
			95	1	1.000	0.633	0.644	0.154
				2	1.000	0.829	0.844	0.245
		0.5	90	1	0.924	0.545	0.545	0.030
				2	0.924	0.685	0.685	0.075

Table 4
Simulated sizes/powers for the four log-rank tests with not-rare events ($\lambda_1 = 0.1$, $\alpha = 0.05$)

λ_2	n	p_1	Censoring prop. $1 - p_D$ (%)	Sampling prop. q (%)	T_{full}	SP_n	V_n	T_{sub}
0.1	200	0.3	75	30	0.070	0.051	0.056	0.092
				40	0.070	0.056	0.052	0.071
		0.5	75	30	0.058	0.059	0.063	0.048
				40	0.058	0.062	0.064	0.053
	400	0.3	75	30	0.054	0.048	0.054	0.066
				40	0.054	0.050	0.049	0.057
		0.5	75	30	0.059	0.052	0.053	0.048
				40	0.059	0.063	0.057	0.044
0.15	200	0.3	75	30	0.454	0.245	0.264	0.246
				40	0.454	0.297	0.313	0.291
		0.5	75	30	0.480	0.310	0.325	0.211
				40	0.480	0.339	0.364	0.284
	400	0.3	75	30	0.659	0.388	0.405	0.323
				40	0.659	0.460	0.471	0.363
		0.5	75	30	0.688	0.431	0.457	0.289
				40	0.688	0.518	0.540	0.383
0.2	200	0.3	75	30	0.852	0.523	0.564	0.455
				40	0.852	0.622	0.654	0.555
		0.5	75	30	0.875	0.613	0.644	0.434
				40	0.875	0.690	0.737	0.555
	400	0.3	75	30	0.983	0.758	0.841	0.660
				40	0.983	0.798	0.866	0.762
		0.5	75	30	0.988	0.842	0.879	0.674
				40	0.988	0.901	0.917	0.775

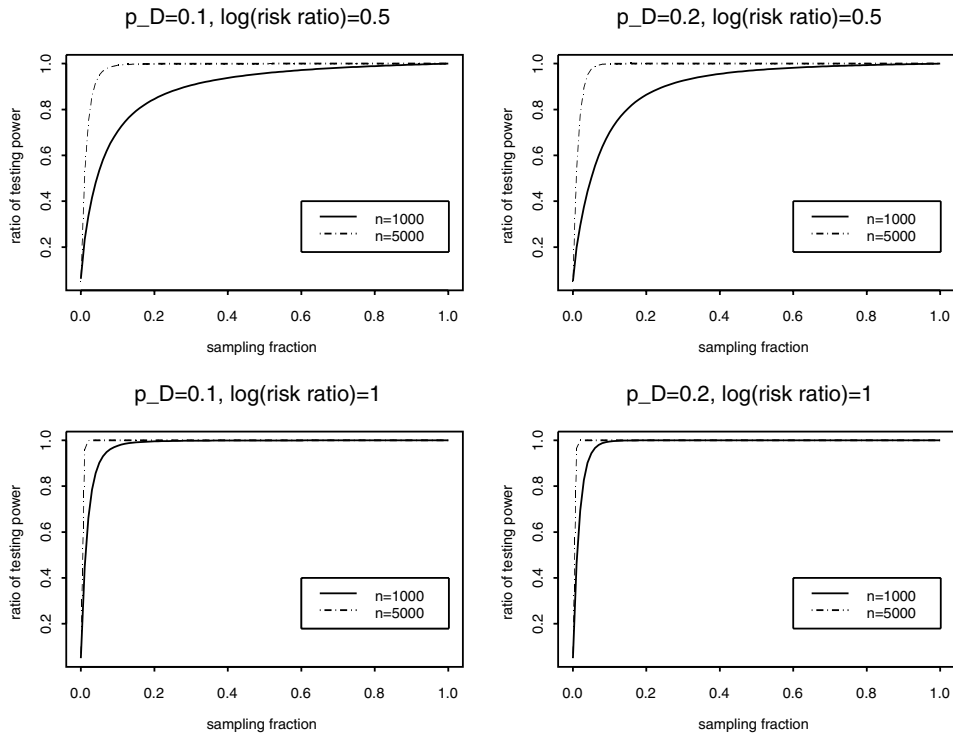


Figure 2. Power of the case-cohort design relative to the full cohort.

4. Practical Implementation and an Example

Based on the approximated power formula in (5), for a given power β , significance level α , the whole cohort size n , the proportion of group j in the whole cohort p_j ($j = 1, 2$), and the failure rate p_D , to detect a hazard ratio of $\exp(\theta)$, the sample size required for the subcohort is

$$\tilde{n} = nBp_D/(n - B(1 - p_D)),$$

$$\text{where } B = (Z_{1-\alpha} + Z_\beta)^2 / (\theta^2 p_1 p_2 p_D). \quad (6)$$

In order for the above formula to be meaningful, the denominator of it needs to be positive, i.e., $n > B(1 - p_D)$. By some manipulation, this is equivalent to

$$\theta > \theta_{lo} \equiv \frac{Z_{1-\alpha} + Z_\beta}{\sqrt{np_1 p_2 p_D}} (1 - p_D)^{\frac{1}{2}}.$$

Note that since the failure rate p_D is usually very small for the case-cohort studies, $1 - p_D \approx 1$. Hence $\theta_{lo} \approx (Z_{1-\alpha} + Z_\beta) / (np_1 p_2 p_D)^{1/2}$, which is the log-hazard ratio that can be detected with the entire cohort. This implies that the case-cohort design will not be able to detect the hazard ratio smaller than the one which can be detected by using the entire cohort. This is a reasonable restriction.

In practice, one might wish to compare the cost efficiency between the test using the case-cohort design and the test using a random sample design, assuming that the group proportions (p_1, p_2), the failure proportion p_D , and the log-hazard ratio θ are constant. The cost efficiency can be measured by the ratio of the sample sizes needed with the random sample design and with the case-cohort design for achieving the same power β ; or equivalently, it is the ratio of the costs between these two designs when the cost of collecting

data for each subject is assumed to be the same. From the power calculation, the number of subjects with the simple random sample design is given by $((Z_{1-\alpha} + Z_\beta) / (\theta(p_1 p_2 p_D)^{1/2}))^2$; while with a case-cohort design, the number of subjects needed is equal to the size of the subcohort plus the additional failures, which is $\tilde{n} + (1 - q)p_D \tilde{n} / q = ((q + (1 - q)p_D)^2 / q) ((Z_{1-\alpha} + Z_\beta) / (\theta(p_1 p_2 p_D)^{1/2}))^2$. Therefore, the cost efficiency is given by $r = q / (q + (1 - q)p_D)^2$. When $r > 1$, the sample size needed for the case-cohort design is smaller than that using simple random sample design. The larger r values ($r > 1$) correspond to better savings using the case-cohort design when the failure rate is small. In the top panel of Figure 3, we plot the cost efficiency as a function of the sampling proportion when failure rate is 5%, 10%, and 20%, respectively. We can see that when the failure rate is very small (5% and 10%), there is an optimal sampling proportion, which is the one corresponding to the peak of the curve. Not much cost efficiency is gained when sampling proportion is larger than 50%. In the bottom panel of Figure 3, we plot the cost efficiency region of $r > 1$ as a function of the sampling and the failure proportion. It is clear that the sample size needed for the case-cohort design is smaller when failure rate is less than 20% for most of the sampling frame.

For illustrative purposes, we consider the Atherosclerosis Risk in Communities (ARIC) study. ARIC is a prospective population-based cohort study of the etiology and natural history of atherosclerosis and the etiology of clinical atherosclerotic disease in four U.S. communities (ARIC Investigators, 1989). A total of 15,792 (n) participants completed a home interview and clinic examinations and were prospectively followed for the development of coronary heart disease (CHD) and other vascular events. A genetic polymorphism of platelet

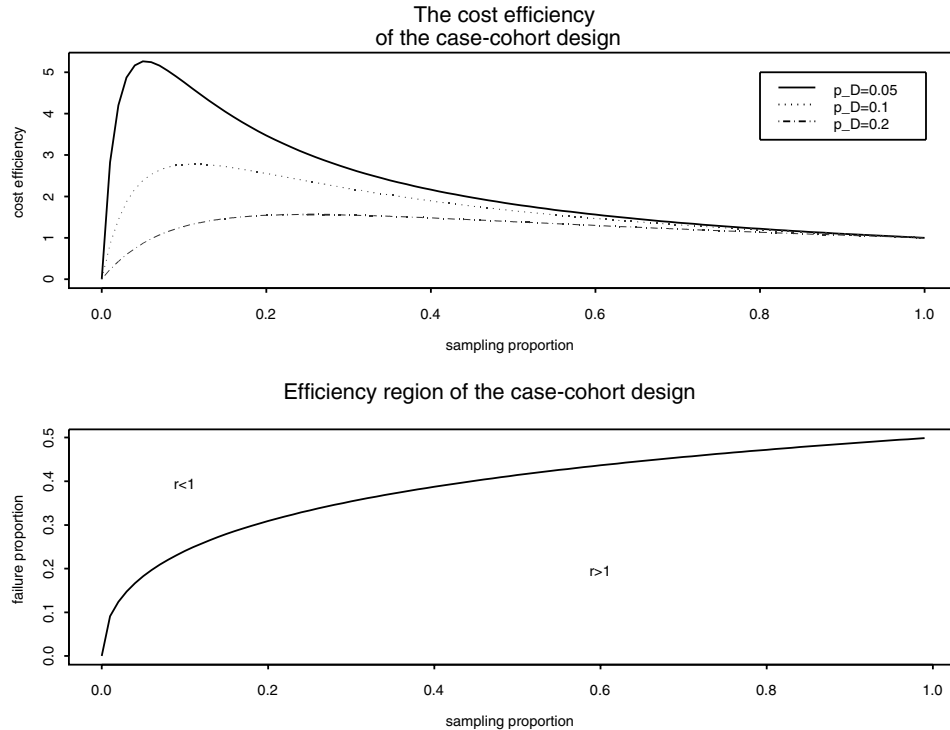


Figure 3. Cost efficiency of the case-cohort design relative to the full cohort.

PI^{A2} allele has been recently proposed as a potential genetic factor linking to platelet hyperaggregability and increased risk of myocardial infarction and therefore it is of interest to examine the relationship between platelet PI^{A2} allele and incidence of CHD in the ARIC study. Since it is expensive to analyze the PI^{A1/A2} genotypes and the CHD incidence rate is low ($p_D = 3\%$) in the ARIC study, a case-cohort design was desirable. Based on the results of Batalla et al. (1999), we use the prevalence of PI^{A2} allele carriers to be 25% (p_1). Based on (6), with a test size of 0.05 and power of 80%, to detect a hazard ratio of 1.35, 1.40, or 1.45, the required subcohort size is 1463, 722, or 468, respectively.

5. Conclusion

We have described two log-rank type test statistics for data from a case-cohort design. Explicit formulas are obtained for the calculation of power and sample size. Our simulation studies show that when the incidence of disease is low in a large cohort, a case-cohort design, which samples a small fraction of the cohort, can produce fairly high power.

The two log-rank type test statistics, SP_n and V_n , can be considered as a modified log-rank test applied to the case-cohort design. Both of these tests are nonparametric. Our theoretical justification and numerical study indicate that when event rate is rare, these two tests give similar powers, while V_n has higher power when the event rate is moderate sized. Thus, we recommend to use V_n when event rate is moderate sized while either SP_n or V_n can be used when event rate is small.

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RÉSUMÉ

En épidémiologie et dans les essais de médecine préventive, on s'intéresse fréquemment à l'influence des facteurs de risques sur la maladie étudiée. Si cette maladie est rare, et que le coût du recueil de l'information individuelle sur les facteurs de risque est important, il est alors utile, dans certaines études, de recourir à un plan en cas-contrôle: c'est-à-dire qu'on sélectionne l'ensemble des sujets malades, auxquels on adjoit un petit échantillon tiré au sort parmi les sujets non atteints. Alors qu'il existe une large bibliographie sur la façon d'analyser ces données de type cas-contrôle, peu de travaux, en revanche, se sont penchés sur le problème de la taille de l'effectif. Dans cet article, nous décrivons deux tests applicables au cas-contrôle, tests qui peuvent être vus comme une généralisation naturelle du test du log-rank tel qu'on l'utilise sur une cohorte complète. De ces deux tests, nous dérivons, sous une forme explicite, un calcul de puissance et d'effectif. Des simulations illustrent l'efficacité de ces tests en cas-contrôle, et l'on présente aussi un exemple d'utilisation de la formule.

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APPENDIX

Power Formula for SP_n

Let $Q_j(t) = P(X_j \geq t)$, $S_j(t) = P(T_j \geq t)$, and $\pi_j(t) = P(C_j \geq t)$. Self and Prentice (1988) show that under H_0 when $S_1(t) = S_2(t) = S(t)$, $n^{-1/2} SP_n$ weakly converges to a normal distribution with mean zero and variance given by

$$\begin{aligned} \sigma_{sp}^2 &= \sigma^2 + \int \int \frac{w(t)}{p_1\pi_1(t) + p_2\pi_2(t)} \frac{w(s)}{p_1\pi_1(s) + p_2\pi_2(s)} \\ &\quad \times Q(t, s) \frac{dS(s)}{S(s)} \frac{dS(t)}{S(t)} \\ &= \sigma^2 + \frac{2p_1p_2(1-q)}{q} \\ &\quad \times \int \frac{w(t)\pi_1(t)\pi_2(t)}{p_1\pi_1(t) + p_2\pi_2(t)} \int_0^t w(s) \frac{dS(s)}{S(s)} dS(t), \end{aligned}$$

where $q = \lim \tilde{n}/n$, σ^2 is the asymptotic variance of the log-rank test from the full cohort and

$$\begin{aligned} Q(t, s) &= \frac{1-q}{q} p_1 p_2 S(t \vee s) \\ &\quad \times (p_1 \pi_1(t) \pi_1(s) \pi_2(t \vee s) + p_2 \pi_2(t) \pi_2(s) \pi_1(t \vee s)). \end{aligned}$$

In particular, when $w(t) = 1$,

$$\sigma_{sp}^2 = \sigma^2 + \frac{2p_1p_2(1-q)}{q} \int \frac{\pi_1(t)\pi_2(t)}{p_1\pi_1(t) + p_2\pi_2(t)} S(t) \Lambda(t) d\Lambda(t).$$

Thus, to estimate the variance σ_{sp}^2 , one can first obtain a consistent estimate of σ^2 using

$$\hat{\sigma}^2 = \frac{1}{n} \int \frac{\tilde{Y}_2(t)^2 d\tilde{N}_1(t) + \tilde{Y}_1(t)^2 d\tilde{N}_2(t)}{(\tilde{Y}_1(t) + \tilde{Y}_2(t))^2}.$$

Then after noting that $\frac{1}{n} \tilde{Y}_j(t) \approx p_j \pi_j(t) S(t)$ and that

$$\tilde{\Lambda}(t) = \int_0^t \frac{d\tilde{N}_1(t) + d\tilde{N}_2(t)}{\tilde{Y}_1(t) + \tilde{Y}_2(t)} \approx \Lambda(t)/q,$$

a consistent estimate for σ_{sp}^2 can be given by

$$\begin{aligned} \hat{\sigma}_{sp}^2 &= \hat{\sigma}^2 + \frac{1-\hat{q}}{n} \int \int \frac{w(t)w(s)}{(\tilde{Y}_1(t) + \tilde{Y}_2(t))(\tilde{Y}_1(s) + \tilde{Y}_2(s))} \\ &\quad \times (\tilde{Y}_1(t)\tilde{Y}_1(s)\tilde{Y}_2(t \vee s) + \tilde{Y}_2(t)\tilde{Y}_2(s)\tilde{Y}_1(t \vee s)) d\tilde{\Lambda}(t) d\tilde{\Lambda}(s) \\ &= \hat{\sigma}^2 + \frac{1-\hat{q}}{n} \sum_{j,j'=1}^2 \sum_{i=1}^{n_j} \sum_{i'=1}^{n_{j'}} \\ &\quad \times \left[\frac{\Delta_{ij} \Delta_{i'j'} w(X_{ij}) w(X_{i'j'})}{(\tilde{Y}_1(X_{ij}) + \tilde{Y}_2(X_{ij}))^2 (\tilde{Y}_1(X_{i'j'}) + \tilde{Y}_2(X_{i'j'}))^2} \right. \\ &\quad \times (\tilde{Y}_1(X_{ij}) \tilde{Y}_1(X_{i'j'}) \tilde{Y}_2(X_{ij} \vee X_{i'j'}) \\ &\quad \left. + \tilde{Y}_2(X_{ij}) \tilde{Y}_2(X_{i'j'}) \tilde{Y}_1(X_{ij} \vee X_{i'j'})) \right], \end{aligned}$$

which is equivalent to the expression in (4).

To obtain the power formula under the simplification conditions (a)–(c) and $w(t) = 1$, first, note that the asymptotic

expectation of $n^{-1/2} SP_n$ under H_A is the same as the asymptotic expectation of the usual log-rank test statistic (1) under H_A and can be approximated by $n^{-1/2} \int \frac{\tilde{Y}_1(t)\tilde{Y}_2(t)}{\tilde{Y}_1(t) + \tilde{Y}_2(t)} (d\Lambda_1(t) - d\Lambda_2(t)) \approx \tilde{n}^{-1/2} (q)^{1/2} \theta p_2 D_1$, where D_j , $j = 1, 2$, denote the total number of failures in group j . Second, we want to simplify the following expression in $\hat{\sigma}_{sp}^2$:

$$\begin{aligned} &2 \sum_{j=1}^2 \sum_{i=1}^{n_j} \left\{ \frac{\Delta_{ij} w(X_{ij}) \tilde{Y}_1(X_{ij}) \tilde{Y}_2(X_{ij})}{(\tilde{Y}_1(X_{ij}) + \tilde{Y}_2(X_{ij}))^2} \right. \\ &\quad \times \left[\sum_{j'=1}^2 \sum_{i'=1}^{n_{j'}} \frac{\Delta_{i'j'} w(X_{i'j'}) I(X_{i'j'} \leq X_{ij})}{\tilde{Y}_1(X_{i'j'}) + \tilde{Y}_2(X_{i'j'})} \right] \Big\} \\ &- \sum_{j=1}^2 \sum_{i=1}^{n_j} \frac{\Delta_{ij} w(X_{ij}) \tilde{Y}_1(X_{ij}) \tilde{Y}_2(X_{ij})}{(\tilde{Y}_1(X_{ij}) + \tilde{Y}_2(X_{ij}))^3}. \end{aligned} \quad (A.1)$$

To do that, we order all the failure times from the smallest to the largest by using different indexes $k = 1, 2, \dots$, and denote the corresponding risk set size for group j in the full cohort as m_k^j , $m_k^j, \dots$, $j = 1, 2$. Since the size of risk set in the subcohort is about q times the size of risk set in the full cohort, the above expression can be approximated by

$$\begin{aligned} &\frac{1}{q} \left[2 \sum_k p_1 p_2 \sum_{k' \leq k} \frac{1}{m_{k'}^1 + m_{k'}^2} - \sum_k p_1 p_2 \frac{1}{m_k^1 + m_k^2} \right] \\ &= \frac{1}{q} p_1 p_2 2 \sum_k \frac{d_k - 1/2}{m_k^1 + m_k^2}, \end{aligned}$$

where d_k is the number of failure times greater than or equal to the k th failure time. From the assumption (b) that failures are very few, we have $m_k^1 + m_k^2 \approx n$, so (A.1) is approximately $p_1 p_2 (D_1 + D_2)^2 / (nq)$. Furthermore, since it is known that $\hat{\sigma}^2$, which estimates the asymptotic variance for the usual log-rank test, is approximately $(p_2^2 D_1 + p_1^2 D_2)/n$, the estimate $\hat{\sigma}_{sp}^2$ can be approximated by $(p_2^2 D_1 + p_1^2 D_2)/n + (1-q)p_1 p_2 (D_1 + D_2)^2 / (n^2 q)$. Therefore, the power of our test at H_A is approximated by

$$\Phi \left(Z_\alpha + \frac{\tilde{n}^{-1/2} \theta p_2 D_1 \sqrt{q}}{\sqrt{(p_2^2 D_1 + p_1^2 D_2)/n + (1-q)p_1 p_2 (D_1 + D_2)^2 / (n^2 q)}} \right).$$

Finally, since $n = \tilde{n}/q$ and $D_j \approx p_j p_D n$, where p_D is the failure proportion, the power is approximated by the expression in (5).

Power Formula for V_n

The derivation of the asymptotic distribution of V_n can follow the same lines for SP_n in Self and Prentice (1988). Eventually, it can be shown that under H_0 , $n^{-1/2} V_n$ has an asymptotically normal distribution with mean zero and variance

$$\begin{aligned} \sigma_V^2 &= \sigma^2 + \int \int \frac{w(t)}{p_1\pi_1(t) + p_2\pi_2(t)} \frac{w(s)}{p_1\pi_1(s) + p_2\pi_2(s)} \\ &\quad \times Q^*(t, s) \frac{dS(s)}{S(s)} \frac{dS(t)}{S(t)}, \end{aligned}$$

where σ^2 is the asymptotic variance of the log-rank test using the full cohort and

$$Q^*(t, s) = \frac{1-q}{q} \\ \times \left\{ p_2^2 \pi_2(t) \pi_2(s) [p_1 P(T \geq C_1 \geq t \vee s) \right. \\ \left. - p_1^2 P(T \geq C_1 \geq t) P(T \geq C_1 \geq s)] \right. \\ \left. + p_1^2 \pi_1(t) \pi_1(s) [p_2 P(T \geq C_2 \geq t \vee s) \right. \\ \left. - p_2^2 P(T \geq C_2 \geq t) P(T \geq C_2 \geq s)] \right. \\ \left. + p_1^2 p_2^2 \pi_1(t) \pi_2(s) P(T \geq C_1 \geq s) P(T \geq C_2 \geq t) \right. \\ \left. + p_1^2 p_2^2 \pi_2(t) \pi_1(s) P(T \geq C_1 \geq t) P(T \geq C_2 \geq s) \right\}.$$

For estimating the variance σ_V^2 , we define $Y_j^c(t) = \frac{n}{n} \sum_{i=1}^{n_j} I(X_{ij} \geq t, \Delta_{ij} = 0, \xi_{ij} = 1), j = 1, 2$. Since $\frac{1}{n} Y_j^*(t) \approx p_j \pi_j(t) S(t)$ and $\frac{1}{n} Y_j^c(t) \approx p_j P(T \geq C_j \geq t)$, a consistent estimator for σ_V^2 is

$$\hat{\sigma}_V^2 = \hat{\sigma}^2 \\ + \frac{1-\hat{q}}{n\hat{q}} \int \int \frac{w(t)w(s)Y_2^*(t)Y_2^*(s)}{(Y_1^*(t) + Y_2^*(t))(Y_1^*(s) + Y_2^*(s))} \\ \times \left[Y_1^c(t \vee s) - \frac{1}{n} Y_1^c(t) Y_1^c(s) \right] \\ \times \frac{d\bar{N}_1(t) + d\bar{N}_2(t)}{Y_1^*(t) + Y_2^*(t)} \frac{d\bar{N}_1(s) + d\bar{N}_2(s)}{Y_1^*(s) + Y_2^*(s)}$$

$$+ \frac{1-\hat{q}}{n\hat{q}} \int \int \frac{w(t)w(s)Y_1^*(t)Y_1^*(s)}{(Y_1^*(t) + Y_2^*(t))(Y_1^*(s) + Y_2^*(s))} \\ \times \left[Y_2^c(t \vee s) - \frac{1}{n} Y_2^c(t) Y_2^c(s) \right] \\ \times \frac{d\bar{N}_1(t) + d\bar{N}_2(t)}{Y_1^*(t) + Y_2^*(t)} \frac{d\bar{N}_1(s) + d\bar{N}_2(s)}{Y_1^*(s) + Y_2^*(s)} \\ + \frac{1-\hat{q}}{n^2\hat{q}} \int \int \frac{w(t)w(s)Y_1^*(t)Y_2^*(s)Y_1^c(s)Y_2^c(t)}{(Y_1^*(t) + Y_2^*(t))(Y_1^*(s) + Y_2^*(s))} \\ \times \frac{d\bar{N}_1(t) + d\bar{N}_2(t)}{Y_1^*(t) + Y_2^*(t)} \frac{d\bar{N}_1(s) + d\bar{N}_2(s)}{Y_1^*(s) + Y_2^*(s)} \\ + \frac{1-\hat{q}}{n^2\hat{q}} \int \int \frac{w(t)w(s)Y_1^*(s)Y_2^*(t)Y_1^c(t)Y_2^c(s)}{(Y_1^*(t) + Y_2^*(t))(Y_1^*(s) + Y_2^*(s))} \\ \times \frac{d\bar{N}_1(t) + d\bar{N}_2(t)}{Y_1^*(t) + Y_2^*(t)} \frac{d\bar{N}_1(s) + d\bar{N}_2(s)}{Y_1^*(s) + Y_2^*(s)}.$$

It is clear that under rare event assumption, both test statistics, SP_n and V_n , are approximately the same. Using similar arguments as in deriving the power formula for SP_n , it can be shown that the power for the alternative hypothesis $H_A: \Lambda_2(t) = e^\theta \Lambda_1(t), \theta > 0$, is approximated by the expression in (5).